

Seat No.: _____

Enrolment No. 250103002004

NATIONAL FORENSIC SCIENCES UNIVERSITY

M.Sc. Cyber Security- Semester - II - April-2026

Subject Code: CTMSCS SII P1

Date: 24/04/2026

Subject Name: Network Security

Time: 2.30 PM TO 5.30 PM

Total Marks: 100

Instructions:

1. Write down each question on separate page.
2. Attempt all questions.
3. Make suitable assumptions wherever necessary.
4. Figures to the right indicate full marks.

		Marks
Q.1	Attempt any three.	
(a)	Explain the OSI model and compare it with the TCP/IP model.	08
(b)	Discuss OSI Security architecture in detail.	08
(c)	A university network suddenly experiences slow performance. Investigation reveals excessive broadcast traffic and MAC table overflow in switches. i. Identify the attack. ii. Explain how it works. iii. Suggest countermeasures.	08
(d)	An organization is assigned the IP block 192.168.10.0/24 and needs 5 subnets. i. Calculate subnet mask ii. Number of hosts per subnet.	08
Q.2	Attempt any three.	
(a)	Describe the penetration testing life cycle.	08
(b)	A penetration tester is hired to assess a company. During reconnaissance, they gather employee details via social media and perform WHOIS lookup. i. Identify phases of penetration testing involved. ii. Suggest how the organization could prevent such information leakage.	08
(c)	Explain the functioning of IDS, IPS, and Firewall with suitable diagrams.	08
(d)	Explain Phishing attack and its countermeasures.	08
Q.3	Attempt any three.	
(a)	Explain RSA with examples	08
(b)	Explain OSCAR methodology in detail with example.	08
(c)	Explain symmetric vs asymmetric encryption with examples.	08

(d) A financial institution suffers a suspected data breach. Logs are collected from firewalls, web servers, and database servers, but timestamps are inconsistent due to unsynchronized system clocks. 08

- i. Identify the challenges in correlating evidence.
- ii. Explain how investigators can reconstruct the timeline.
- iii. Suggest preventive measures for future investigations.

Q.4

Attempt any two.

- ✓(a) Explain Active vs Passive Scanning and types of port scanning (TCP/UDP) 07
- ✓(b) Describe Hash Functions and Message Digest. 07
- (c) Explain 802.11 protocols and WLAN architecture. 07

Q.5

Attempt any two.

- ✓(a) Encrypt "ATTACKATDAWN" using key "LEMON". 07
- ✓(b) A company uses WEP encryption and experiences unauthorized access. 07
- a. Explain why WEP is insecure.
- b. Suggest improvements using WPA2.
- ✓(c) An employee is suspected of leaking confidential data. Network logs show unusual data transfers during non-working hours. 07

- Describe how network forensic techniques can confirm the insider threat.
- What types of evidence should be collected?
- Discuss legal and ethical considerations.

END OF PAPER

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NATIONAL FORENSIC SCIENCES UNIVERSITY

Semester End Examination

M.Sc. Cyber Security - Semester – II - April – 2026

Subject Code: CTMSCS SII P2

Date: 28/04/2026

Subject Name: Malware Analysis

Time: 14.30 hrs – 17.30 hrs

Total Marks: 100

Instructions:

1. Read the question paper carefully.
2. Write down each question on a separate page.
3. Attempt all questions.
4. Make suitable assumptions wherever necessary.
5. Figures to the right indicate full marks.

	Marks
Q.1	
Attempt any three.	
(a) Convert the following C code to appropriate assembly with respect to reverse engineering.	08

```
#include <stdio.h>

void main() {
    int password = 1234;
    int input;
    int attempts = 0;

    while (attempts < 3) {
        printf("Enter password: ");
        scanf("%d", &input);

        if (input == password) {
            printf("Access Granted!\n");
            break;
        } else {
            printf("Wrong Password!\n");
        }

        attempts++;

        if (attempts == 3) {
            printf("Too many failed attempts. Access Denied.\n");
        }
    }
}
```

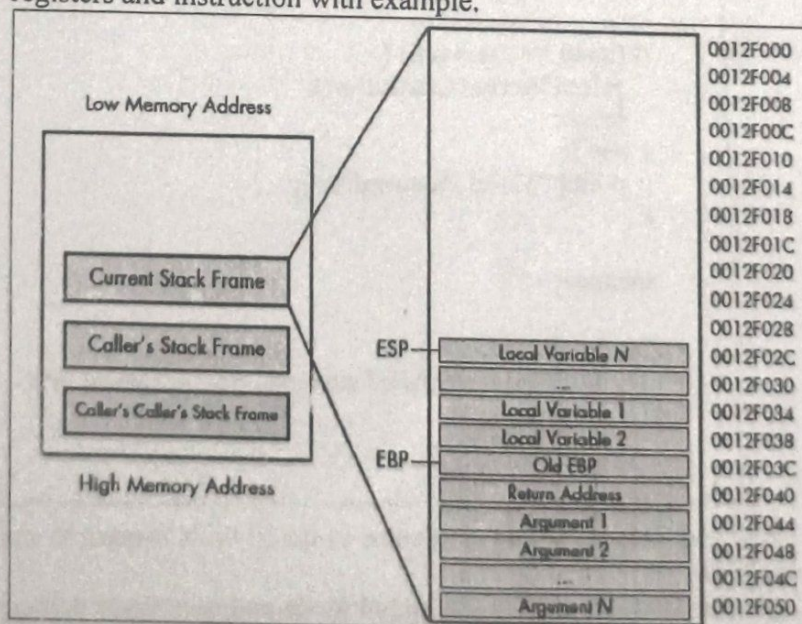
- | | | |
|------|---|----|
| ✓(b) | Draw and discuss the PE structure in detail with respect to malware analysis. | 08 |
| ✓(c) | State the differences between kernel mode and user mode debugger. | 08 |
| ✓(d) | Omkar is has recently joined as malware analyst in Perox, one good day he was analyzing the given sample and identified use of VirtualAllocEx and WriteProcessMemory windows APIs. What type of possible technique the malware is using here? Discuss with justification. | 08 |

- Q.2 Attempt any three. 08
- (a) Explain the functions and features of ollydbg/immunity debugger. Write a case study where you can change the execution of the program with the help of debugger. (Write logic's pseudocode for explanation) 08
- (b) Write a detailed note on basic dynamic analysis technique with tools and examples. 08
- (c) Discuss the packing and unpacking in detail. 08
- (d) Explain the real-world case study of the malware attack in detail. 08

- Q.3 Attempt any three. 08
- (a) Write a detailed note on live memory analysis with volatility for malware analysis. 08
- (b) How RAT and botnet types of malwares behaves? Discuss them technically with its key differences. 08
- (c) Write a detailed note on anti-reverse engineering techniques used by the malwares. 08
- (d) Discuss the linux malware analysis with appropriate example. 08

- Q.4 Attempt any two. 07
- (a) What is used to pause the execution of the binary in the debugger? Explain that in detail. 07
- (b) What can be observed and identified from the given disassembly snippet in fig. A at page no. 3? Justify in detail. 07
- (c) Explain the reverse engineering procedure of the android malware with appropriate technical example. 07

- Q.5 Attempt any two. 07
- (a) Illustrate the given image and also explain the relevant assembly registers and instruction with example. 07



- (b) Draw and explain the android architecture in detail. 07
- (c) Write a detailed note on various malware analysis techniques with their merit and demerits. 07

.text:00401037	lpAddress	= dword ptr -8
.text:00401037	hObject	= dword ptr -4
.text:00401037	lpFileName	= dword ptr 8
.text:00401037		
.text:00401038	push	ebp
.text:0040103A	mov	ebp, esp
.text:0040103D	sub	esp, 10h
.text:00401044	mov	[ebp+lpAddress], 0
.text:00401046	push	0 ; hTemplateFile
.text:0040104B	push	80h ; dwFlagsAndAttributes
.text:0040104D	push	3 ; dwCreationDisposition
.text:0040104F	push	0 ; lpSecurityAttributes
.text:00401051	push	1 ; dwShareMode
.text:00401056	push	80000000h ; dwDesiredAccess
.text:00401059	mov	eax, [ebp+lpFileName]
.text:0040105A	push	eax ; lpFileName
	call	ds:CreateFileA

.text:00401149	lea	eax, [ebp+StartupInfo]
.text:0040114C	push	eax ; lpStartupInfo
.text:0040114D	push	0 ; lpCurrentDirectory
.text:0040114F	push	0 ; lpEnvironment
.text:00401151	push	4 ; dwCreationFlags
.text:00401153	push	0 ; bInheritHandles
.text:00401155	push	0 ; lpThreadAttributes
.text:00401157	push	0 ; lpProcessAttributes
.text:00401159	push	0 ; lpCommandLine
.text:0040115B	mov	ecx, [ebp+lpApplicationName]
.text:0040115E	push	ecx ; lpApplicationName
.text:0040115F	call	ds:CreateProcessA

Fig. A

--- End of Paper---

Seat No.: _____

Enrolment No. 250103002004

NATIONAL FORENSIC SCIENCES UNIVERSITY

M. Sc. Cyber Security - Semester - II - April-2026

Subject Code: CTMSCS SII P3

Date: 27.04.2026

Subject Name: Mobile Security

Time: 2.30 PM TO 5.30 PM

Total Marks: 100

Instructions:

1. Write down each question on separate page.
2. Attempt all questions.
3. Make suitable assumptions wherever necessary.
4. Figures to the right indicate full marks.

Q.1	Attempt any three.	Marks
✓(a)	Explain the android architecture in detail with an illustrative diagram. Discuss each layer with suitable examples.	08
(b)	A mobile application allows access to restricted data without authentication. Identify the type of vulnerability and explain how you would test it.	08
✓(c)	Discuss about Reverse Engineering using Apktool and JADX.	08
✓(d)	Explain the concept of android application security vulnerability assessment using QARK.	08
Q.2	Attempt any three.	
✓(a)	Describe the exploitation of Android application vulnerabilities with respect to the following	08
✓(a)	Insecure login (2 Marks)	
✓(b)	Hard core issues (2 Marks)	
✓(c)	Insecure data storage (2 Marks)	
✓(d)	Input validation issues (2 Marks)	
✓(b)	Explain the concept of Android File System with supported file formats.	08
✓(c)	Explain the phases of penetration testing methodology in mobile application with suitable example.	08
(d)	Explain the concept of Frida and describe its role in Android dynamic instrumentation for application security analysis.	08
Q.3	Attempt any three.	
✓(a)	Describe the stages of android boot process and explain each stage in detail.	08
(b)	Discuss about security auditing using Drozer and explain the types of security auditing approaches.	08
✓(c)	Explain traffic analysis for Android devices and describe the difference between passive and active analysis methods.	08
(d)	Discuss the OWASP Mobile Top 10 vulnerabilities in detail with prevention methods.	08

Q.4

Attempt any two.

- ✓(a) Explain the Android Debug Bridge (ADB) and illustrate the syntax and usage of important ADB commands. 07
- (b) Discuss the role of Hex Dump and Dex Dump command in Reverse Engineering. 07
- ✓(c) Write short notes on: 07
- Traffic interception tools (3 Marks)
 - SSL pinning (2 Marks)
 - Proxy-based analysis (2 Marks)

Q.5

Attempt any two.

- ✓(a) Discuss about MobSF (Mobile Security Framework) and explain its static and dynamic analysis features in detail. 07
- (b) Describe the importance of secure source code review in Android security. 07
- ✓(c) Discuss the components of Android Application along with the foreground and background process in detail. 07

END OF PAPER

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Enrollment No: 250103002004

NATIONAL FORENSIC SCIENCES UNIVERSITY

M. Sc. Cyber Security (Semester – II)
Semester End Examination - April 2026

Subject Code: CTMSCS SII P4

Subject Name: Incident Response and Digital Forensics

Time: 02:30 PM to 05:30 PM

Date: 30/04/2026

Total Marks: 100

Instructions:

1. Write down each question on separate page.
2. Attempt all questions.
3. Use of Calculator is permitted.
4. Figures to the right indicate full marks.

- | | Marks |
|---|-------|
| Q.1 Answer the following (Attempt Any Three) | |
| ☒ (a) i) Define cybersecurity incident. How does it differ from an event? Explain with real-world examples. | 08 |
| ☒ ii) Explain the goals and benefits of having an Incident Response Plan (IRP) in an organization. | |
| ☒ (b) Your department head has called an urgent meeting to discuss the security of client information and you being a security consultant he has asked you to explain the different data classification techniques to the team. | 08 |
| ☒ (c) Scenario:
An employee at a financial firm receives a seemingly internal email from the HR department asking them to urgently verify their personal details through a web link. A few others report similar emails, and one of them accidentally enters their credentials. Hours later, IT detects unusual login attempts to the email server from multiple foreign IPs using that _____ user's _____ credentials. | 08 |
| ☒ i) Identify and list at least two precursors and two indicators from this scenario. | |
| ☒ ii) How could these precursors, if detected earlier, have prevented the incident? | |
| ☒ iii) Differentiate precursors and indicators. | |
| (d) Briefly discuss on Incident prioritization and what are the factors used to prioritize an Incident. | 08 |
| Q.2 Answer the following (Attempt Any Three) | |
| ☒ (a) Scenario:
On 16 th December 2025 at 03:00 PM, a financial company experiences unauthorized access to its internal payroll system. Logs show suspicious login attempts from an unknown external IP, followed by privilege escalation. The incident was detected by the SOC, and containment actions were carried out within 2 hours. Based on the scenario above, prepare a short incident response documentation containing all the required components from incident summary to lessons learned. | 08 |
| ☒ (b) Scenario:
During an ongoing cyber incident, you suspect that a critical server has been compromised and is actively communicating with an attacker's system. The team faces a decision: disconnect the server from the network immediately (containment) or keep it | 08 |

online a bit longer to monitor attacker activity and gather further evidence. Discuss the factors that influence this containment decision, and explain how you would approach making the choice.

08

- (c) After an incident has been resolved and normal operations are restored, explain what steps the incident response team should take in the post-incident phase and why each is important. Provide at least two specific examples of post-incident activities and describe how each activity contributes to improving future incident response or preventing similar incidents.

- (d) An organization experiences two security events at the same time:
- (i) A virus is detected and automatically quarantined on an employee's laptop containing only public data;
 - (ii) A database containing customers' personal information may have been accessed by an unauthorized external party. Discuss how will you prioritize the incident and explain which event should be treated as a higher severity security incident and why.

08

Q.3 Answer the following (Attempt Any Three)

- (a) Your SOC team is training new analysts on log sources and their relevance.
- i) Explain any three major log sources used in incident response. For each log sources, mention what type of attack or behavior it helps in detecting for the organization.
 - ii) Consider this Log format:

08

```
203.0.113.45 - - [15/Mar/2024:16:11:42 +0530] "GET /product?id=10 UNION SELECT username,password FROM users HTTP/1.1" 500 1024 "-" "Mozilla/5.0"
```

Explain in detail the parameters/elements of this log & also try to identify the potential attack technique.

- (b) What is a file system? Discuss NTFS file system and explain log collection approaches on Windows OS.

08

- (c) Your organization plans to improve visibility into internal network traffic and build a Security Operations Centre (SOC). As an Incident Response (IR) analyst, explain the essential components of a SOC and illustrate a high-level SOC architecture.

08

- (d) Scenario:

08

A cloud-based HR portal used by your organization recently experienced unauthorized access attempts. An employee reported that salary records were altered, and several login attempts were made from unknown IP addresses. Additionally, the portal became temporarily unavailable during business hours due to excessive failed login attempts. Explain how each component of the CIA triad was impacted in this scenario.

Q.4 Answer the following (Attempt Any Two)

- (a) Scenario:

07

Your SOC team handled a complex insider threat incident. The team deployed 4 staff members to resolve it. The incident was one of 10 handled that month. Use the data below to estimate the Cost Per Incident using the standard method of cost estimation.

Staff	Monthly Salary	Time Spent (%)
Augustine	₹90,000	20%
Birju	₹70,000	30%

Camphor	₹60,000	20%
Dave	₹70,000	30%

Other Details:

- Capital Expenditure: ₹30,000; Overhead cost: 10%; Throughput: 10 incidents

i) Calculate the staff cost for each team member and then compute the total staff cost for the incident and calculate the cost per incident.

ii) Suggest two ways the organization could reduce the cost per incident based on your analysis.

(b) During an enterprise-wide malware outbreak, an IR team begins a live response on multiple systems. 07

(i) List and explain six categories of artifacts you would collect for further analysis.

(ii) Discuss the risks associated with performing live response.

✓(c) Explain the typical structure and functions of a Security Operations Centre (SOC). In your answer, also describe the roles and responsibilities of Incident Response Team Members. 07

Q.5 Answer the following (Attempt Any Two)

✓(a) Scenario: 07

During an investigation, as an investigator you came across web browsers and email clients. Kindly list and explain what evidences you will collect from i) Web Browsers & ii) Email Clients and why.

(b) Explain: 07

i) Windows Registry

ii) System Event Logs

iii) Prefetch Files

✓(c) Explain with Examples: 07

i) AAA

ii) Access Control Techniques

--- End of Paper---